In [ ]:
```
#STEP-1: Importing the libraries
```

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

In [ ]:
```
#STEP-2: Loading the data
```

In [6]:
```
df = pd.read_csv('Netflix_Data.csv')
print(df.head())
```

```
  Show_id     Type                  Title         Director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                Cast        Country  \
0                                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...            NaN
3                                                NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India

           Date_added  Release_year Rating   Duration  \
0  September 25, 2021          2020  PG-13     90 min
1  September 24, 2021          2021  TV-MA  2 Seasons
2  September 24, 2021          2021  TV-MA   1 Season
3  September 24, 2021          2021  TV-MA   1 Season
4  September 24, 2021          2021  TV-MA  2 Seasons

                                           Listed_in  \
0                                      Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
3                             Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

                                         Description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...
```

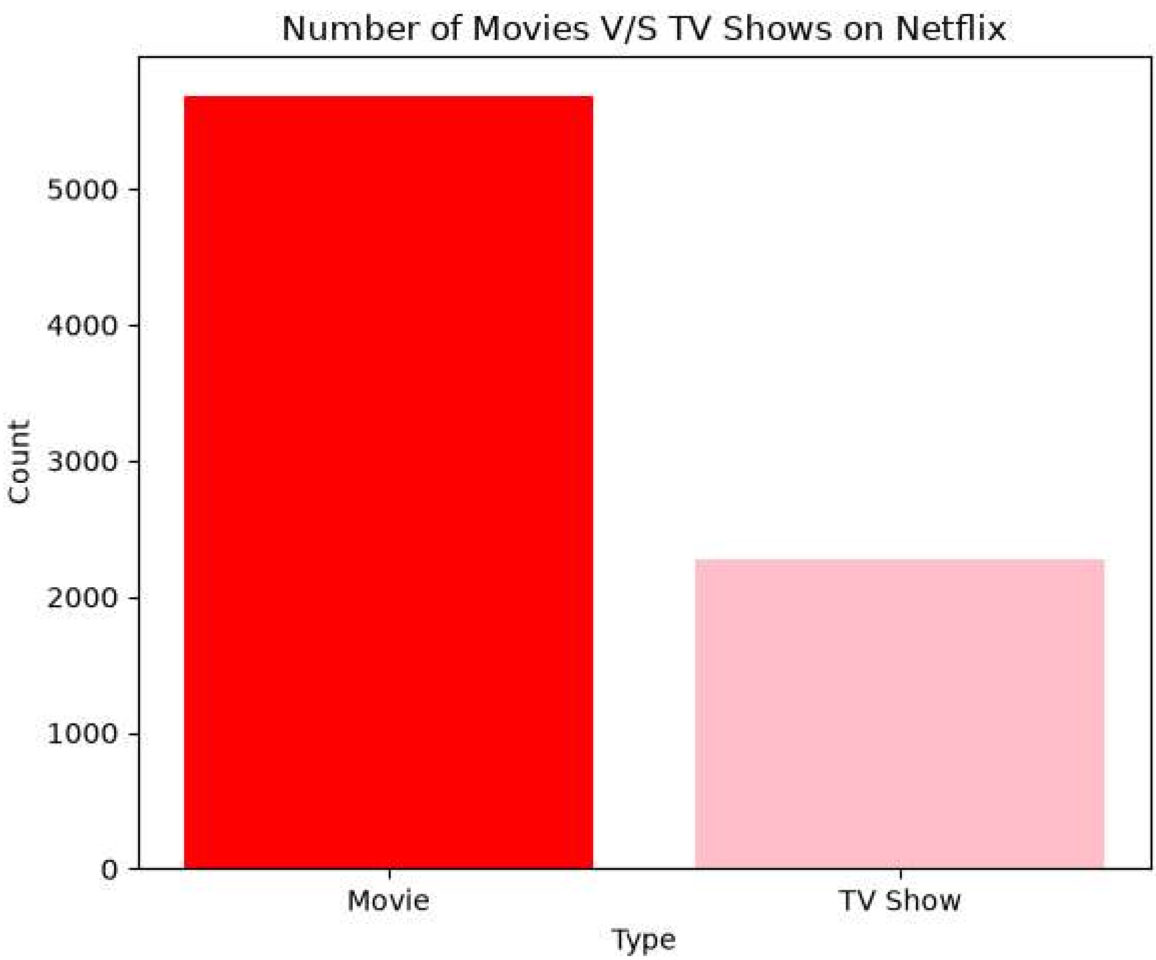
In [6]:
```
#STEP-3: Cleaning the data
```

In [7]:
```
df = df.dropna(subset=['Type', 'Release_year','Rating', 'Country', 'Duration'])
```

In [ ]:
```
#STEP-4: Visualization
```

In [58]:
```
'''
BAR CHART:- Movies V/S TV shows
```

```
'''
type_counts = df['Type'].value_counts()
plt.figure(figsize = (6,5))
plt.bar(type_counts.index, type_counts.values, color=['red', 'pink'])
plt.title('Number of Movies V/S TV Shows on Netflix')
plt.xlabel('Type')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('Movies_vs_TVshows.png')
plt.show()
```

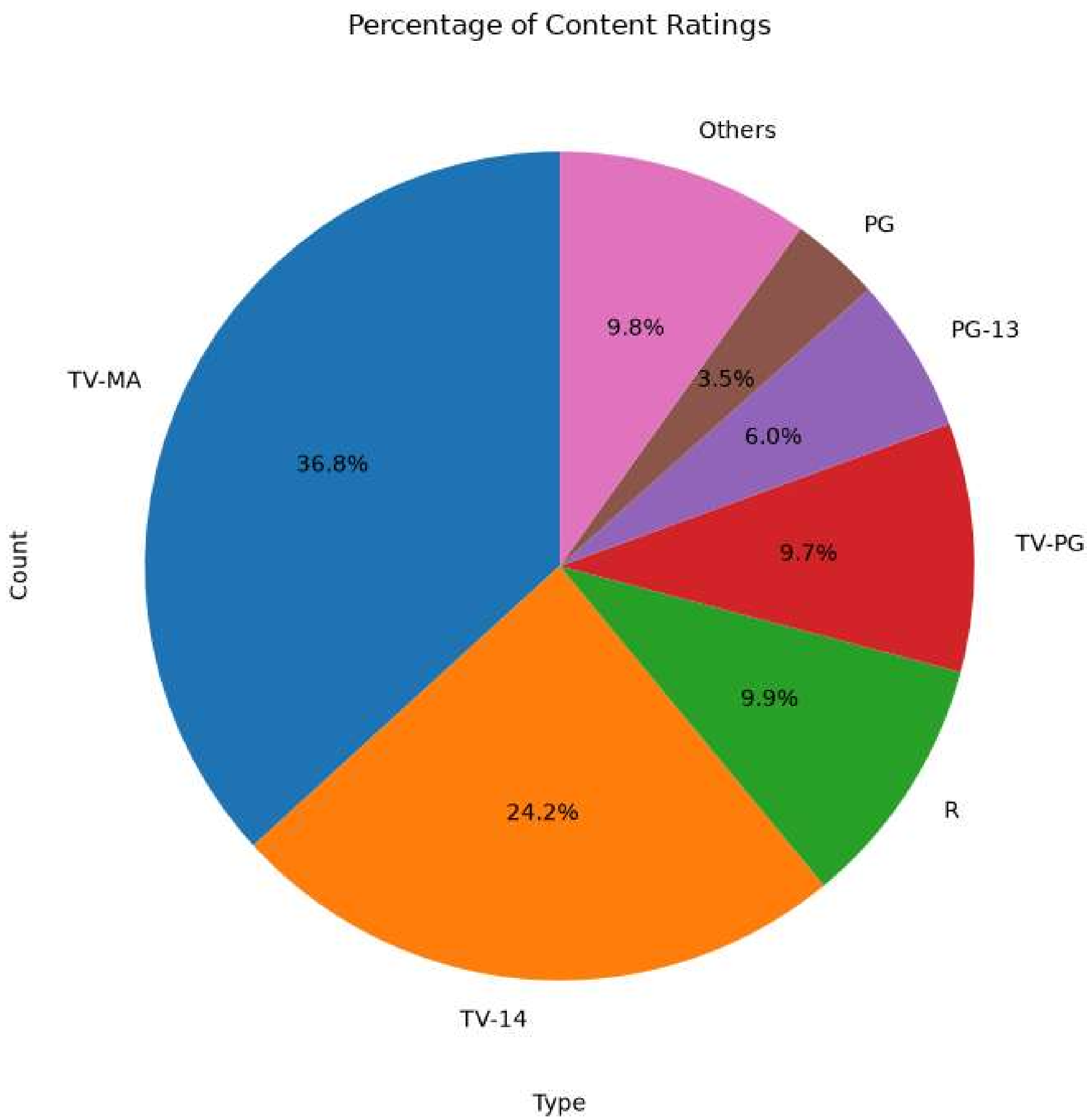


In [ ]:
```
'''Conclusion'''
Movies are much more numerous than TV Shows in the Netflix dataset.
```

In [ ]:

In [42]:
```
'''
PIE CHART:- Content Rating Distribution
'''
rating_counts = df['Rating'].value_counts()
top = rating_counts[:6]
others = rating_counts[6:].sum()
top["Others"] = others
plt.figure(figsize = (7,7))
plt.pie(top, labels = top.index, autopct = '%1.1f%%', startangle = 90)
plt.title('Percentage of Content Ratings')
```

```
plt.xlabel('Type')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('Content_Ratings_pie.png')
plt.show()
```

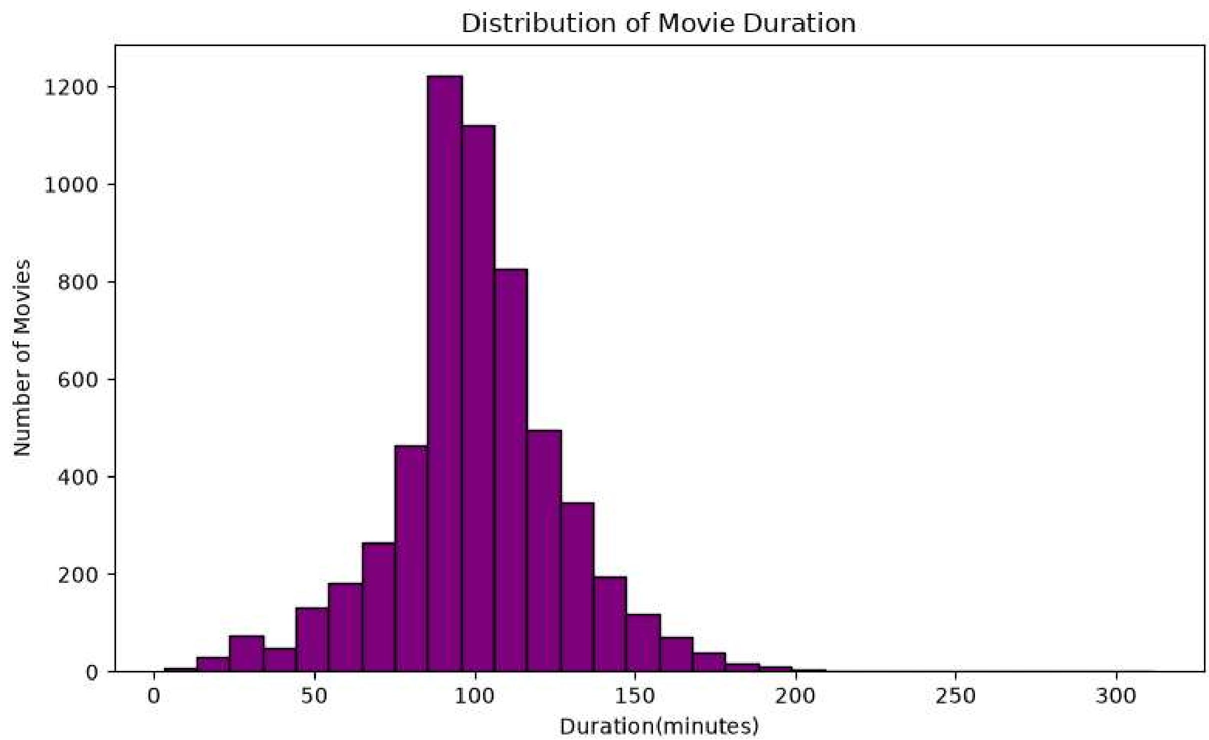


In [ ]:
```
'''Conclusion'''
TV-MA(36.8%) is the largest category while PG-13(6.0%) and PG(3.5%) accounts for sm
```

In [ ]:

In [57]:
```
'''
HISTOGRAMS:- Movie Duration Distribution
'''
movie_df = df[df['Type'] == 'Movie'].copy()
movie_df['duration_int'] = movie_df['Duration'].str.replace('min','').astype(int)

plt.figure(figsize=(8,5))
```

```
plt.hist(movie_df['duration_int'], bins=30, color='purple', edgecolor = 'black')
plt.title('Distribution of Movie Duration')
plt.xlabel('Duration(minutes)')
plt.ylabel('Number of Movies')
plt.tight_layout()
plt.savefig('Movie_Duration_histogram.png')
plt.show()
```

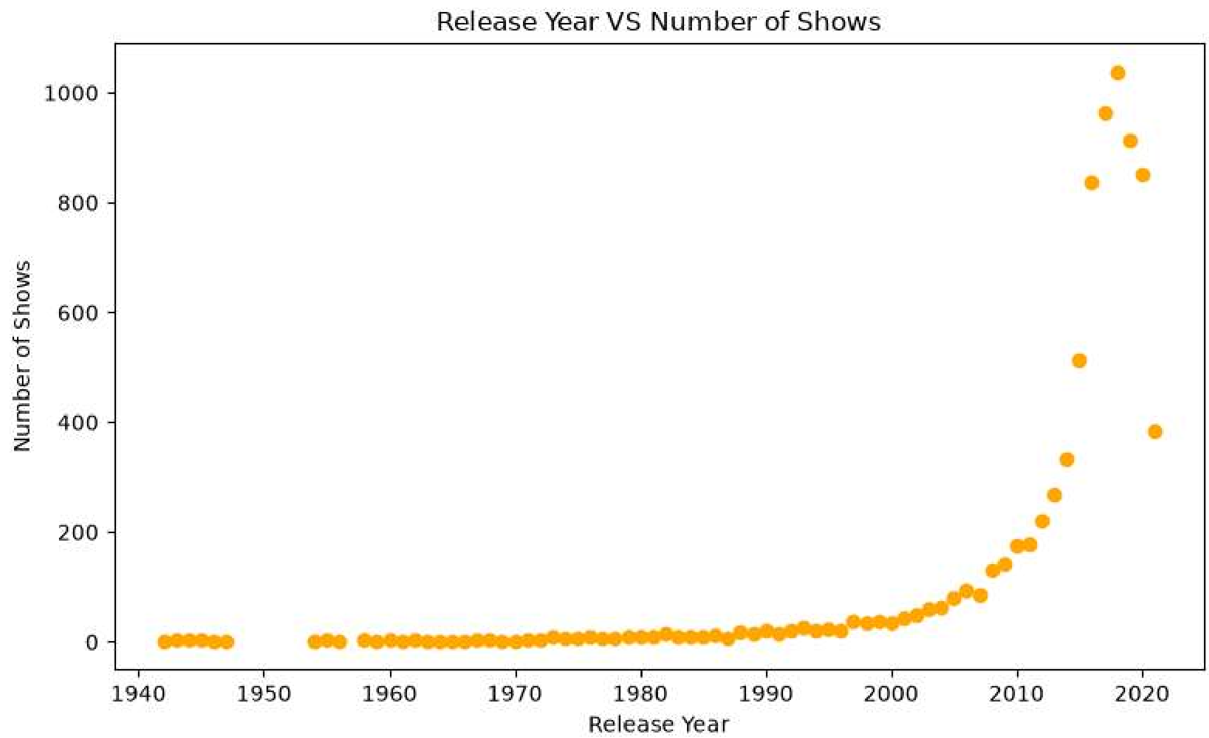


In [ ]:
```
'''Conclusion'''
Most Netflix movies are concentrated around 90-120 minutes which indicates Netflix
```

In [ ]:

In [ ]:
```
'''
SCATTER PLOT:- Release Year V/S Number of Shows
'''
```

In [55]:
```
release_counts = df['Release_year'].value_counts().sort_index()
plt.figure(figsize=(8,5))
plt.scatter(release_counts.index, release_counts.values, color = 'orange')
plt.title('Release Year VS Number of Shows')
plt.xlabel('Release Year')
plt.ylabel('Number of Shows')
plt.tight_layout()
plt.savefig('Release_Year_scatter.png')
plt.show()
```

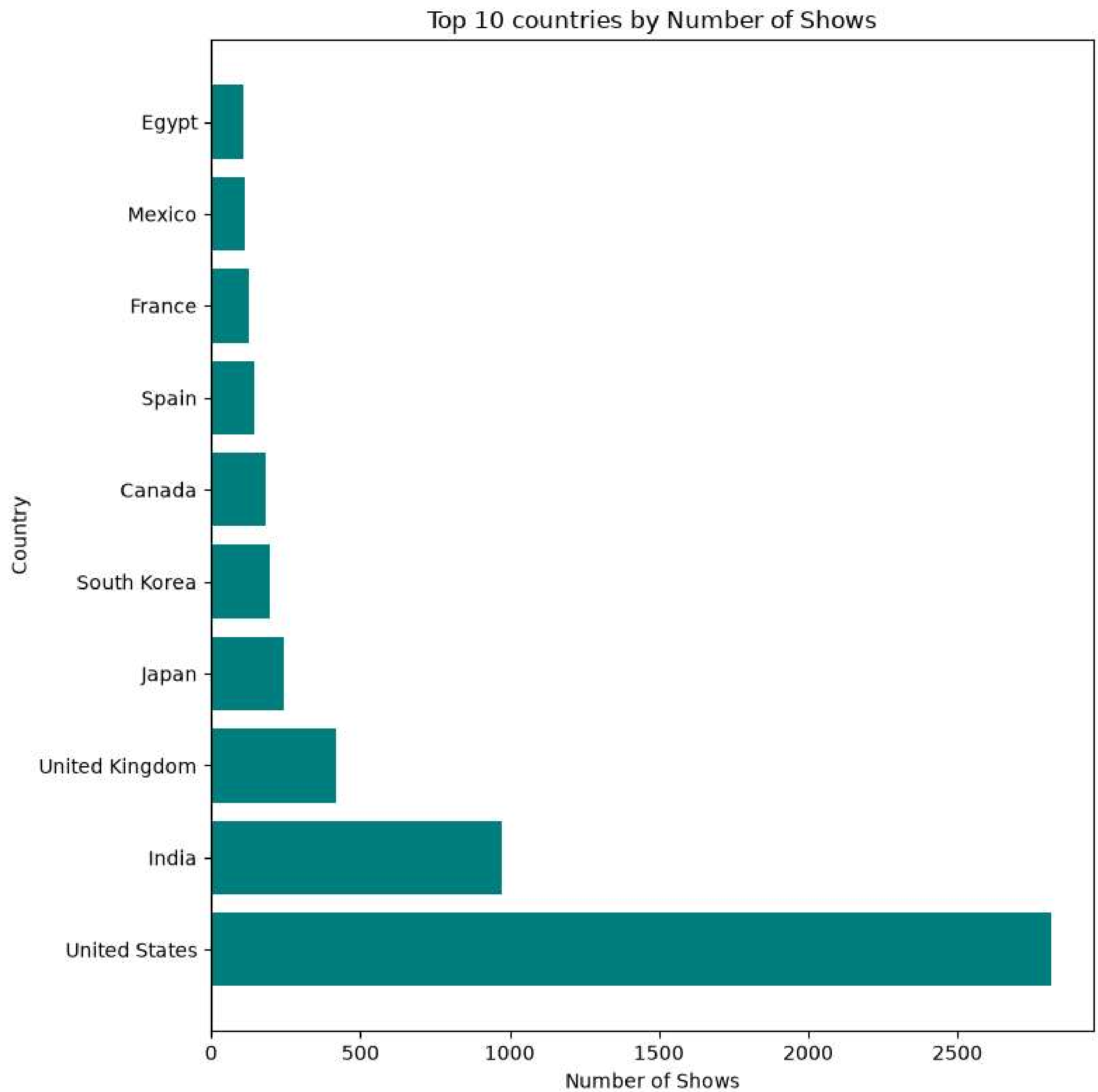


In [ ]:
```
'''Conclusion'''
The number of shows increases significantly in more recent years, indicating that N
```

In [ ]:
```

```

In [ ]:
```
'''
BAR GRAPH: Top 10 countries shows
'''
```

In [65]:
```
country_counts = df['Country'].value_counts().head(10)
plt.figure(figsize=(8,8))
plt.barh(country_counts.index, country_counts.values, color = 'teal')
plt.title('Top 10 countries by Number of Shows')
plt.xlabel('Number of Shows')
plt.ylabel('Country')
plt.tight_layout()
plt.savefig('Top 10_Countries.png')
plt.show()
```



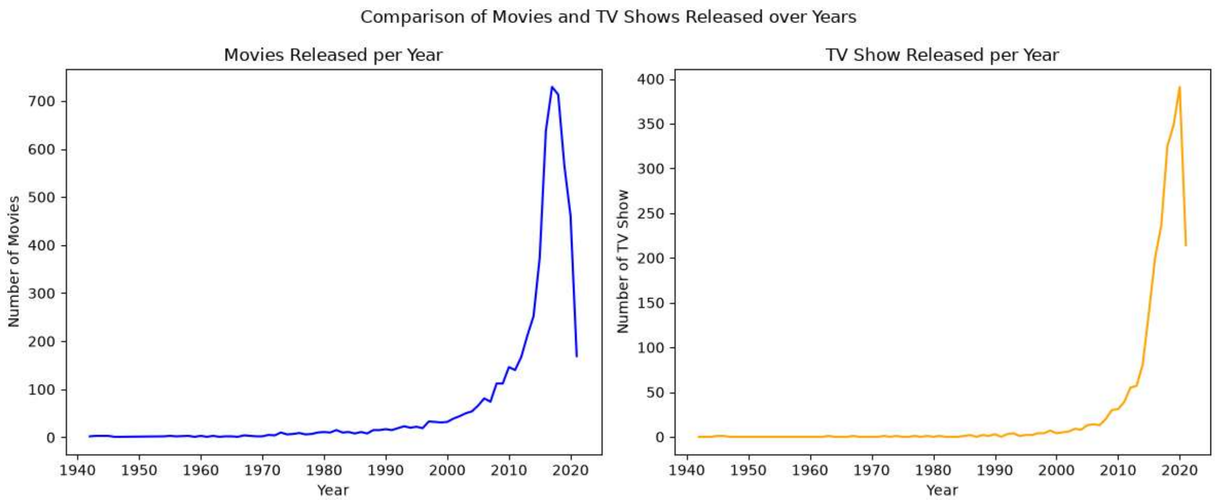
In [ ]:
```
'''Conclusion'''
The United States has the highest number of Netflix title, dominating the catalogue
```

In [ ]:
```

```

In [ ]:
```
'''
SUBPLOT:- Movie V/S TV Shows by Year
'''
```

In [12]:
```
content_by_year = df.groupby(['Release_year', 'Type']).size().unstack().fillna(0)

fig, ax = plt.subplots(1,2, figsize=(12,5))

#First subplot:movies
ax[0].plot(content_by_year.index, content_by_year['Movie'], color = 'blue')
ax[0].set_title('Movies Released per Year')
ax[0].set_xlabel('Year')
ax[0].set_ylabel('Number of Movies')
```

```
#Second subplot:TV shows
ax[1].plot(content_by_year.index, content_by_year['TV Show'], color = 'orange')
ax[1].set_title('TV Show Released per Year')
ax[1].set_xlabel('Year')
ax[1].set_ylabel('Number of TV Show')

fig.suptitle('Comparison of Movies and TV Shows Released over Years')

plt.tight_layout()
plt.savefig('Movie_TV_Shows_Comparison.png')
plt.show()
```



In [ ]:
```
'''Conclusion'''
Both show an increasing trend over the years, with a significant rise in recent yea
```